FORD VEV

Proving Ground Operational System

Visao Geral do Projeto

Documentacao Tecnica - ABNT NBR 14724

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Sistema integrado de telemetria, gestão de turnos e analytics para o Campo de Provas Ford Tatuí (TPG). Combina um aplicativo PWA para coleta de dados em campo (Operadores) com um dashboard Streamlit para análise gerencial (Coordenação/Engenharia).

1 Architecture Overview

```
+--------------------------------------------------------------+
|                    FORD VEV ECOSYSTEM                        |
|                                                              |
|  +----------------------+    +--------------------------+   |
|  |   DATA COLLECTION    |    |   ANALYTICS DASHBOARD    |   |
|  |   (PWA / Firebase)   |    |   (Streamlit + Plotly)   |   |
|  |                      |    |                          |   |
|  |  - Shift tracking    |◄--►|  - KPI dashboards       |   |
|  |  - GPS telemetry     |    |  - Live monitoring       |   |
|  |  - Technical reports |    |  - Fuel analysis         |   |
|  |  - Braking/accel     |    |  - Vehicle health cards  |   |
|  |  - Fuel refueling    |    |  - Pilot rankings        |   |
|  |  - Route logging     |    |  - Anomaly detection     |   |
|  |  - PDF generation    |    |  - Export to Excel/Sheets|   |
|  +---------+------------+    +-----------+--------------+   |
|            |                             |                   |
|            ▼                             ▼                   |
|  +--------------------------------------------------------+  |
|  |              FIREBASE BACKEND                          |  |
|  |                                                        |  |
|  |  +----------+  +-----------+  +------------------+    |  |
|  |  |Firestore |  |Realtime   |  |Auth (email/pass) |    |  |
|  |  |(historic)|  |Database   |  |@ford.com only    |    |  |
|  |  |          |  |(live ops) |  |with auto-approval|    |  |
|  |  +----------+  +-----------+  +------------------+    |  |
|  +--------------------------------------------------------+  |
|                                                              |
|  +--------------------------------------------------------+  |
|  |   EXTERNAL INTEGRATIONS                                |  |
|  |   - Google Sheets (via Apps Script)                    |  |
|  |   - Microsoft Forms (project-specific checklists)      |  |
|  |   - EmailJS (approval workflows)                       |  |
|  |   - Gemini API (laudo text refinement)                 |  |
|  +--------------------------------------------------------+  |
+--------------------------------------------------------------+
```

2 Repo Structure

```
ford-vev-global-main/
|
+-- index.html                  # PWA shell - 3 screens (Home/Telemetry/History)
+-- app.js                      # Main app logic - shift engine, UI, laudos (~4K lines)
+-- auth-security.js            # Firebase Auth, EmailJS, approvals, audit logs
+-- turno-engine.js             # Shift lifecycle, GPS, fuel, pre-cadastro (~3K lines)
+-- laudo-ia.js                 # Technical report creation, Gemini integration
+-- analytics.js                # Analytics dashboard (modal inside PWA)
+-- vev-improvements.js         # Accessibility, modals, performance patches
+-- dados-mestres.js            # Master data - projects, track tests, vehicles
+-- forms-data.js               # Microsoft Forms registry per project
+-- forms-engine.js             # Dynamic forms modal renderer
+-- notificacoes-engine.js      # In-app notification system (Firestore)
|
+-- service-worker.js           # PWA service worker (cache-v9)
+-- manifest.json               # PWA manifest (standalone, pt-BR)
+-- offline.html                # Offline fallback page
|
+-- style.css                   # Full design system (~6K lines)
+-- style-ws.css                # WebKit scrollbar styles
|
+-- package.json                # Vite + Firebase v10
+-- vite.config.js              # Build: minify, copy static JS to dist/
+-- firebase.json               # Hosting config (dist/, Cache-Control, cleanUrls)
+-- .firebaserc                 # Default project: ford-vev
|
+-- dashboard/
|   +-- app.py                  # Streamlit dashboard - 10 tabs (~2.8K lines)
|   +-- ford_vev_colab.py       # Google Colab companion script
|   +-- ford_vev_colab.ipynb    # Jupyter notebook version
|   +-- requirements.txt        # Python deps
|   +-- .streamlit/config.toml  # Theme: Ford dark blue + cyan accent
|
+-- Jsonfirebase/
|   +-- gestao-de-veiculos-municipal-*.json  # Service account key
|
+-- Script vev tabela.txt       # Google Apps Script - syncs to Google Sheets
|
+-- docs/                       # This documentation
    +-- README.md
    +-- DATA_COLLECTION.md
    +-- ANALYTICS_DASHBOARD.md
    +-- DATABASE.md
    +-- DEPLOYMENT.md
```

3 Quick Start

3.1 Data Collection App (PWA)

```
# Install dependencies
npm install

# Start Vite dev server
npm run dev
# Opens at http://localhost:5173

# Build for production
npm run build

# Deploy to Firebase Hosting
npm run deploy
```

3.2 Analytics Dashboard

```
cd dashboard

# Create virtual environment (recommended)
python -m venv .venv
.venv\Scripts\activate

# Install dependencies
pip install -r requirements.txt

# Run Streamlit
python -m streamlit run app.py --server.headless true
# Opens at http://localhost:8501
```

4 Firebase Project

Property	Value
Project ID	\`ford-vev\`
Auth Domain	\`ford-vev.firebaseapp.com\`
RTDB URL	\`https://ford-vev-default-rtdb.firebasei
Storage Bucket	\`ford-vev.firebasestorage.app\`
Hosting Site	\`ford-vev.web.app\`

5 Key Numbers

Metric	Value
JS files (modules)	10 (copied verbatim to dist/)
HTML lines	4,272 (3 screens, 17+ modals)
CSS lines	~6,180 (custom design system)
Dashboard tabs	10 (Streamlit)

Firestore collections	19
RTDB paths	9
Service cache version	`ford-vew-cache-v9`
Precached assets	14

6 Dependencies

Frontend: Firebase v10 (compat SDK via CDN), jsPDF, Chart.js, Leaflet (lazy), EmailJS, Google Fonts (Inter + Material Icons)

Build: Vite 5, ESLint, Prettier

Dashboard: Streamlit 1.31, Pandas 2.1, Plotly 5.18, NumPy 1.26, SciPy 1.12, Folium 0.16, gspread 6.0, OpenPyXL 3.1

7 Skills & Agents (openforge)

Custom skills for AI-assisted development are defined in .opencode/:

- power-bi-analysis - DAX measures, star schema, Firebase connector for Power BI
- dashboard-optimization - Caching, session state, error boundaries, skeleton loading
- frontend-ux - Progressive disclosure, card patterns, responsive grid
- data-analysis-visual - Fleet benchmarking, health scores, anomaly detection
- refresh-ux - User-controlled refresh, state preservation

8 License

Internal Ford Motor Company - TPG Insight AI